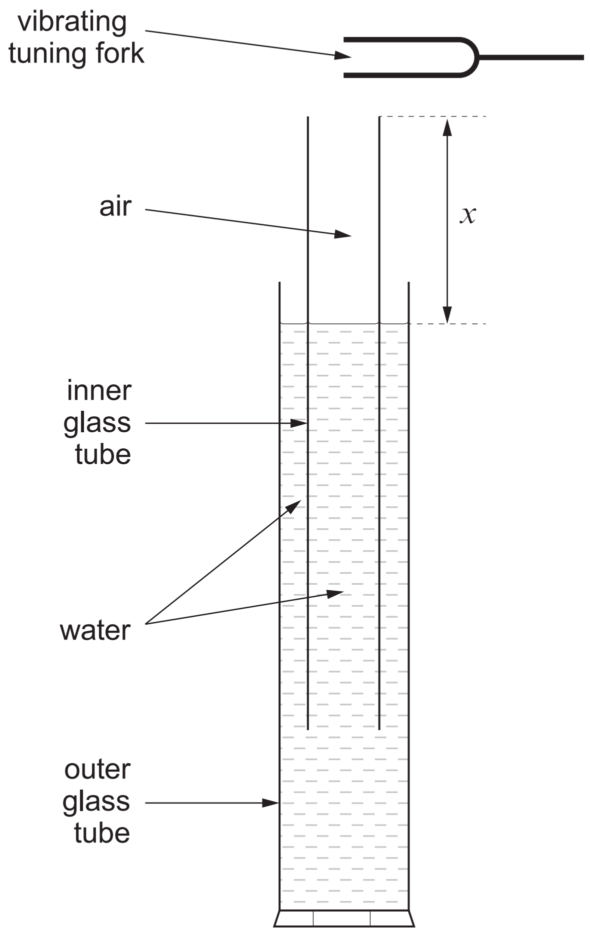
6. Matthew uses the apparatus shown to find a value for the speed of sound in air using stationary waves.



Starting with a very low value of distance x, he gradually raises the inner glass tube. At a certain value of x, the note of the tuning fork is heard loudly. Matthew then measures and records x.

He repeats the procedure several times, obtaining these results.

x / m	0.327	0.329	0.322	0.328	0.332	0.323

The tuning fork frequency, f, is given as 256.0 Hz ± 1%.

Examiner only

(a) The speed of sound in air, v_s, can be found from the equation:

$$v_s = 4fx$$

(i) Calculate the mean value of x along with its **percentage** uncertainty. [2]

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(ii) Hence calculate the value of v_s along with its **absolute** uncertainty. [2]

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Examiner only

(b) The loud sound is due to stationary waves of frequency, f, in the air column. There is a node of displacement at the water surface and an antinode of displacement close to the open end (top) of the inner glass tube.

(i) Derive the equation: [2]

$$v_s = 4fx$$

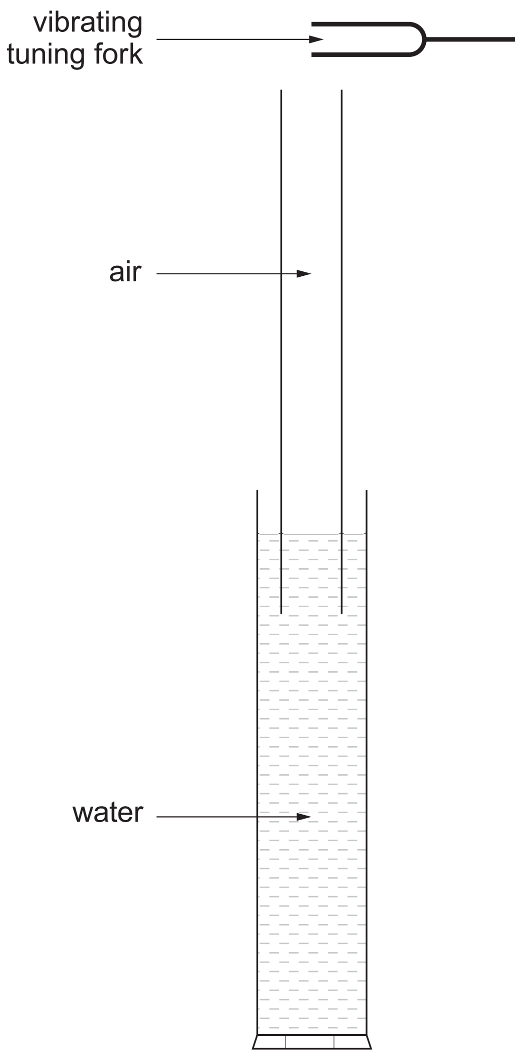
....

....

....

....

(ii) If the inner glass tube is raised considerably further a second distance is found at which a loud note of frequency, f, is heard. This again corresponds to a stationary wave in the air column. Show the positions of its nodes and antinodes of displacement **on the diagram below**. [2]



8